Adaptive Diet Recommendation System for Patients with Both Type 2 Diabetes and Cancer

Problem

Patients with type 2 diabetes (T2D) undergoing cancer treatment must balance blood glucose management with adequate nutritional intake as symptoms change. Between consultations, they need practical support adapting everyday food choices to changing appetite, eating difficulties, and nutritional needs.

Customer Segments

Our primary users are patients managing T2D during cancer treatment at home. Family caregivers support food selection and preparation. Patients and caregivers are potential subscription payers; clinical dietitians are potential referral and evaluation partners.

Early Adopters

Our initial users are patients with T2D receiving outpatient cancer treatment who experience changing appetite or eating difficulties, lack timely support for adjusting food choices, and are willing to report symptoms independently or with caregiver assistance.

Existing Alternatives

Alternatives include dietitian consultations, written dietary guidance, and general meal planners such as Eat This Much. Our focus is adapting food recommendations between consultations as symptoms change, while considering both diabetes management and nutritional needs during cancer treatment.

Unique Value Proposition

Our service helps patients and caregivers choose foods as symptoms change, combining diabetes management considerations with nutritional needs during cancer treatment. Each recommendation explains how reported conditions affect a food's ranking, making the guidance easier to understand.

High-Level Concept

An adaptive nutrition navigator for patients with T2D and cancer.

Solution

Our rule-based engine ranks foods using reported blood glucose, BMI, appetite, cancer type, and treatment stage. It updates recommendations when inputs change and explains relevant rules. We plan independent dietitian review of recommendation appropriateness across patient scenarios.

Key Metrics

We will track weekly update completion, four-week retention, reported recommendation use, paid conversion, and independent dietitian ratings of recommendation appropriateness.

Exploratory health measures include glucose trends, unintentional weight change, and eating difficulties.

Sustainable Advantage

Our assets combine literature-informed rules, food annotations, and datasets documenting recommendation changes across patient states. We plan to strengthen this knowledge base through independent dietitian review and accumulated evaluation cases, building defensibility through validated quality and continued refinement.

Channels

We plan to recruit initial users through dietitian referrals and outpatient cancer support groups, delivering recommendations through a website. After evaluating usability and recommendation quality, we will seek hospital pilot partnerships and test targeted online advertising.

Cost Structure

Estimated monthly infrastructure costs are $17 for a pilot and $93 for a proposed expansion configuration. Additional costs include development, clinical review, data maintenance, security, support, and customer acquisition. Estimates require validation against usage assumptions.

Revenue Streams

We plan monthly subscriptions paid by patients or family caregivers for adaptive food recommendations and explanations. Pilot research will test willingness to pay, pricing, paid conversion, and retention to assess whether subscription revenue can cover ongoing operating costs.